SentinelSwitch 2.0 Evidence-Driven Network Security & Response Lab

Security posture report

Generated 2026-03-17T21:49:31.284Z · simulated switch-twin clock (dataset day 2026-03-17) · schema 2.0 · SentinelSwitch 2.0.0

Scope and simulation disclaimer. All assets, telemetry, incidents and assessments are synthetic and produced by the SentinelSwitch digital twin. No real network was assessed. Scope: SW01 (WS-C2960X-48FPD-L), 36 inventoried assets, 13 controls, 6 VLANs.

1. Executive summary

209 control assessments were evaluated across 13 controls at the simulated clock. 208 passed, 0 failed, 1 could not be determined and 0 rest on stale evidence. 0 finding(s) remain open (0 high, 0 medium, 0 low); 5 finding(s) were verified closed. 0 incident(s) are open (contained or containment failed); 4 resolved, 4 of them with recovery verified by CTL-RECOVERY-VERIFIED.

ASSESSMENTS	PASS	FAIL	UNKNOWN	STALE	OPEN FINDINGS
209	208	0	1	0	0

2. Assessment counts and methodology

METRIC	VALUE	DEFINITION
coverage	99.5%	share of assessments with a determinate result (PASS or FAIL); UNKNOWN and STALE are excluded from the numerator
pass rate (determinate)	100.0%	PASS divided by (PASS + FAIL); this is control effectiveness among determinate results, not overall compliance
pass rate (all)	99.5%	PASS divided by all assessments; UNKNOWN and STALE count against it

STATE	MEANING
PASS	available evidence demonstrates the condition is satisfied
FAIL	available evidence demonstrates the condition is not satisfied
UNKNOWN	evidence is insufficient to decide; never counted as a pass
STALE	evidence is older than the control's freshness window at the assessment clock; never counted as a pass

Freshness: per-control windows measured against the simulated clock; state-based evidence is read live from the twin and has no window

Results by control

CONTROL	CATEGORY	ASSESSED	PASS	FAIL	UNKNOWN	STALE	STATUS	NIST CSF 2.0
Access port carries its intended VLAN CTL-VLAN-INTENT	Segmentation	36	36	0	0	0	PASS	PR.IR-01, ID.AM-03
802.1X / MAB enforced on user ports CTL-DOT1X-ENFORCED	Access control	27	27	0	0	0	PASS	PR.AA-03, PR.IR-01
Connected endpoints have a verified identity CTL-ENDPOINT-IDENTITY	Identity	28	28	0	0	0	PASS	ID.AM-01, PR.AA-03

Port security matches the port profile CTL-PORT-SECURITY	Access control	35	35	0	0	0	PASS	PR.IR-01, PR.AA-03
Interface configuration matches the baseline CTL-CONFIG-BASELINE	Configuration	50	50	0	0	0	PASS	PR.PS-01
Unused ports are shut down and parked CTL-UNUSED-PORTS	Hardening	12	12	0	0	0	PASS	PR.IR-01, PR.PS-01
DHCP snooping / DAI trust limited to infrastructure trunks CTL-TRUST-BOUNDARY	Segmentation	1	1	0	0	0	PASS	PR.IR-01
Required telemetry sources are current CTL-TELEMETRY-FRESH	Monitoring	4	3	0	1	0	UNKNOWN	DE.CM-01
Endpoint VLANs are mirrored to the sensor CTL-SENSOR-COVERAGE	Monitoring	4	4	0	0	0	PASS	DE.CM-01, ID.AM-03
Containment actions were authorized before execution CTL-RESPONSE-AUTHORIZED	Response	4	4	0	0	0	PASS	RS.MA-01, RS.AN-06
Containment state independently verified CTL-QUARANTINE-VERIFIED	Response	0	0	0	0	0	NOT_ASSESSED	RS.MI-01, RS.AN-06
Restored interfaces returned to their expected state CTL-RECOVERY-VERIFIED	Recovery	4	4	0	0	0	PASS	RC.RP-01, RC.RP-04
Incident evidence bundle is complete CTL-EVIDENCE-COMPLETE	Evidence	4	4	0	0	0	PASS	RS.AN-06, DE.AE-03

3. Control findings

No open findings.

Verified / closed findings

FINDING	CONTROL	ASSET	OPENED	CLOSED	VERIFICATION
FND-CONFIG-BASELINE-Gi1-0-22	Interface configuration matches the baseline	MKT-WS-022	2026-03-17T21:14:00.000Z	2026-03-17T21:15:47.484Z	PASS after remediation · run ASR-20260317T211547-remediation-FND-CONFIG-BASELINE-Gi1-0-22
FND-CONFIG-BASELINE-Gi1-0-35	Interface configuration matches the baseline	ENG-LT-035	2026-03-17T21:14:00.000Z	2026-03-17T21:16:47.484Z	PASS after remediation · run ASR-20260317T211647-remediation-FND-CONFIG-BASELINE-Gi1-0-35
FND-ENDPOINT-IDENTITY-Gi1-0-30	Connected endpoints have a verified identity	CONF-A-TV01	2026-03-17T21:14:00.000Z	2026-03-17T21:17:47.484Z	PASS after remediation · run ASR-20260317T211747-remediation-FND-ENDPOINT-IDENTITY-Gi1-0-30
FND-SENSOR-COVERAGE-vlan-50	Endpoint VLANs are mirrored to the sensor	VLAN 50 QUARANTINE	2026-03-17T21:14:00.000Z	2026-03-17T21:18:47.484Z	PASS after remediation · run ASR-20260317T211847-remediation-FND-SENSOR-COVERAGE-vlan-50

FND-UNUSED-PORTS-Gi1-0-41	Unused ports are shut down and parked	Gi1/0/41	2026-03-17T21:14:00.000Z	2026-03-17T21:19:47.484Z	PASS after remediation · run ASR-20260317T211947-remediation-FND-UNUSED-PORTS-Gi1-0-41

4. Affected assets

No assets currently have open findings.

Unverified assessments (UNKNOWN / STALE)

ASSESSMENT	STATUS	SCOPE	EXPLANATION
ASM-TELEMETRY-FRESH-suricata	UNKNOWN	SEC-TP01	The dataset contains Suricata alerts and flow records but no periodic `stats` events, so sensor liveness cannot be determined from available evidence. Alert absence is not evidence of health.

5. Supporting evidence

Evidence rows for every non-passing assessment and for the incident-related controls. Every assessment in the JSON export carries its full evidence list.

ASSESSMENT	STATUS	EVIDENCE SOURCE	REFERENCE	COLLECTED	DETAIL
ASM-EVIDENCE-COMPLETE-INC-2026-0039	PASS	evidence-bundle	data/incidents/INC-2026-0039	2026-03-17T21:49:36.284Z	11/11 artifacts present
ASM-EVIDENCE-COMPLETE-INC-2026-0040	PASS	evidence-bundle	data/incidents/INC-2026-0040	2026-03-17T21:49:36.284Z	11/11 artifacts present
ASM-EVIDENCE-COMPLETE-INC-2026-0041	PASS	evidence-bundle	data/incidents/INC-2026-0041	2026-03-17T21:49:36.284Z	11/11 artifacts present
ASM-EVIDENCE-COMPLETE-INC-2026-0042	PASS	evidence-bundle	data/incidents/INC-2026-0042	2026-03-17T21:49:36.284Z	11/11 artifacts present
ASM-QUARANTINE-VERIFIED-INC-2026-0042	RETIRED	containment-action	WF-2026-0042-01	2026-03-17T21:14:41.125Z	read-back ok: Access Mode VLAN: 50
ASM-QUARANTINE-VERIFIED-INC-2026-0042	RETIRED	switch-twin	show interfaces Gi1/0/23 switchport	2026-03-17T21:19:47.484Z	Administrative Mode: static access
ASM-RECOVERY-VERIFIED-INC-2026-0039	PASS	containment-action	WF-2026-0039-01	2026-03-17T09:35:10.772Z	restored by tlee: Unmanaged switch removed from conference room; display reconnected directly. Port recovered with shutdown / no shutdown.
ASM-RECOVERY-VERIFIED-INC-2026-0039	PASS	switch-twin	show interfaces Gi1/0/30 switchport	2026-03-17T21:49:36.284Z	Administrative Mode: static access
ASM-RECOVERY-VERIFIED-INC-2026-0039	PASS	switch-twin	show interfaces Gi1/0/30 status	2026-03-17T21:49:36.284Z	Gi1/0/30 CONF-A DISPLAY (gu connected 40 a-full a-1000 10/100/1000BaseTX
ASM-RECOVERY-VERIFIED-INC-2026-0040	PASS	containment-action	WF-2026-0040-02	2026-03-17T09:58:30.514Z	restored by tlee: Unregistered contractor single-board computer removed after physical inspection (ticket SVC-2026-0311). Port restored to service.

ASM-RECOVERY-VERIFIED-INC-2026-0040	PASS	switch-twin	show interfaces Gi1/0/17 switchport	2026-03-17T21:49:36.284Z	Administrative Mode: static access
ASM-RECOVERY-VERIFIED-INC-2026-0040	PASS	switch-twin	show interfaces Gi1/0/17 status	2026-03-17T21:49:36.284Z	Gi1/0/17 OPS SPARE DROP (co notconnect 20 auto auto 10/100/1000BaseTX
ASM-RECOVERY-VERIFIED-INC-2026-0041	PASS	containment-action	WF-2026-0041-04	2026-03-17T11:47:20.288Z	restored by tlee: Vendor disabled connection sharing and left site. Port restored to service.
ASM-RECOVERY-VERIFIED-INC-2026-0041	PASS	switch-twin	show interfaces Gi1/0/31 switchport	2026-03-17T21:49:36.284Z	Administrative Mode: static access
ASM-RECOVERY-VERIFIED-INC-2026-0041	PASS	switch-twin	show interfaces Gi1/0/31 status	2026-03-17T21:49:36.284Z	Gi1/0/31 VENDOR DROP (guest notconnect 20 auto auto 10/100/1000BaseTX
ASM-RECOVERY-VERIFIED-INC-2026-0042	PASS	containment-action	WF-2026-0042-01	2026-03-17T21:49:31.284Z	restored by tlee: Endpoint re-imaged and cleared by IT; returning to production VLAN.
ASM-RECOVERY-VERIFIED-INC-2026-0042	PASS	switch-twin	show interfaces Gi1/0/23 switchport	2026-03-17T21:49:36.284Z	Administrative Mode: static access
ASM-RECOVERY-VERIFIED-INC-2026-0042	PASS	switch-twin	show interfaces Gi1/0/23 status	2026-03-17T21:49:36.284Z	Gi1/0/23 ENG-LT-047 connected 20 a-full a-1000 10/100/1000BaseTX
ASM-RESPONSE-AUTHORIZED-INC-2026-0039	PASS	containment-action	WF-2026-0039-01	2026-03-17T09:12:45.469Z	port_shutdown on Gi1/0/30 applied 09:12:45.469Z
ASM-RESPONSE-AUTHORIZED-INC-2026-0039	PASS	policy-decision	policy tier automatic_containment	2026-03-17T09:12:45.081Z	mode=automatic by policy-engine · Automatic containment approved (severity=high confidence=0.96); action=port_shutdown
ASM-RESPONSE-AUTHORIZED-INC-2026-0040	PASS	containment-action	WF-2026-0040-02	2026-03-17T09:41:06.601Z	port_shutdown on Gi1/0/17 applied 09:41:06.601Z
ASM-RESPONSE-AUTHORIZED-INC-2026-0040	PASS	policy-decision	policy tier automatic_containment	2026-03-17T09:41:06.200Z	mode=automatic by policy-engine · Automatic containment approved (severity=high confidence=0.96); action=port_shutdown
ASM-RESPONSE-AUTHORIZED-INC-2026-0041	PASS	containment-action	WF-2026-0041-04	2026-03-17T11:23:48.244Z	port_shutdown on Gi1/0/31 applied 11:23:48.244Z
ASM-RESPONSE-AUTHORIZED-INC-2026-0041	PASS	policy-decision	policy tier automatic_containment	2026-03-17T11:23:47.848Z	mode=automatic by policy-engine · Automatic containment approved (severity=high confidence=0.95); action=port_shutdown
ASM-RESPONSE-AUTHORIZED-INC-2026-0042	PASS	containment-action	WF-2026-0042-01	2026-03-17T21:14:41.000Z	quarantine_vlan on Gi1/0/23 applied 21:14:41.000Z
ASM-RESPONSE-AUTHORIZED-INC-2026-0042	PASS	policy-decision	policy tier automatic_containment	2026-03-17T21:14:40.585Z	mode=automatic by policy-engine · Automatic containment approved (severity=high confidence=0.97); action=quarantine_vlan

ASM-TELEMETRY-FRESH-suricata	UNKNOWN	suricata	data/suricata/eve.json	2026-03-17T21:49:36.284Z	event types present: alert, flow

6. Incident timeline

INC-2026-0042 · Lateral Reconnaissance — ENG-LT-047 high resolved

Chain: 10.20.0.47 → A4:83:E7:12:34:56 → SW01 → Gi1/0/23 → VLAN 20 → VLAN 50 · opened 2026-03-17T21:14:39.867Z · closed 2026-03-17T21:49:31.284Z by tlee

TIME (UTC)	PHASE	ACTOR	EVENT
2026-03-17 21:14:31.284	telemetry	telemetry	unusual ICMP echo fan-out from 10.20.0.47: 507 addresses across 10.20.0.0/24, 10.30.0.0/24
2026-03-17 21:14:33.507	alert	suricata	Suricata sid 1000010 (medium): SENTINELSWITCH RECON ICMP echo sweep from internal host
2026-03-17 21:14:35.116	telemetry	telemetry	reverse DNS lookup burst from 10.20.0.47 for 10.30.0.0/24 (50 queries/s)
2026-03-17 21:14:36.108	alert	suricata	Suricata sid 1000014 (low): SENTINELSWITCH RECON reverse DNS lookup burst
2026-03-17 21:14:36.874	telemetry	telemetry	TCP connection fan-out from 10.20.0.47 to 26 responding hosts on ports 445, 139, 135, 3389, 22, 5985
2026-03-17 21:14:37.042	alert	suricata	Suricata sid 1000024 (low): SENTINELSWITCH LATERAL WinRM probe from workstation VLAN
2026-03-17 21:14:38.198	alert	suricata	Suricata sid 1000021 (high): SENTINELSWITCH LATERAL SMB probe fan-out to multiple internal hosts
2026-03-17 21:14:38.292	alert	suricata	Suricata sid 1000022 (high): SENTINELSWITCH LATERAL RDP probe fan-out to multiple internal hosts
2026-03-17 21:14:39.600	telemetry	telemetry	anonymous SMB session setup against SRV-FILE01 rejected (STATUS_ACCESS_DENIED)
2026-03-17 21:14:39.649	alert	zeek	Zeek notice Scan::Address_Scan: 10.20.0.47 scanned at least 15 unique hosts on port 445/tcp in 0m2s
2026-03-17 21:14:39.826	alert	suricata	Suricata sid 1000023 (low): SENTINELSWITCH LATERAL SMB session setup with anonymous credentials
2026-03-17 21:14:39.867	detection	sentinel	Rule lateral-reconnaissance matched: severity high, confidence 0.97 (base 0.75; zeek_address_scan+0.10, prior_icmp_sweep+0.07, ptr_burst+0.05)
2026-03-17 21:14:39.950	correlation	sentinel	Resolving actor 10.20.0.47 through DHCP snooping, MAC table and inventory
2026-03-17 21:14:40.092	correlation	sentinel	dhcp_binding [resolved] 10.20.0.47 -> A4:83:E7:12:34:56 (dhcp-snooping, VLAN 20, Gi1/0/23, 23841s remaining)
2026-03-17 21:14:40.346	correlation	sentinel	mac_table [resolved] A4:83:E7:12:34:56 -> SW01 Gi1/0/23 (VLAN 20, DYNAMIC)
2026-03-17 21:14:40.346	correlation	sentinel	interface [resolved] Gi1/0/23 'ENG-LT-047' mode=access access_vlan=20 oper_vlan=20 class=workstation status=connected
2026-03-17 21:14:40.439	correlation	sentinel	identity [resolved] ENG-LT-047 (laptop, Lenovo, Ubuntu 24.04 LTS) criticality=standard user=jchen (Jia Chen, Engineering)
2026-03-17 21:14:40.585	policy	sentinel	Disposition automatic_containment: Automatic containment approved (severity=high confidence=0.97); action=quarantine_vlan
2026-03-17 21:14:40.587	authorization	sentinel	Containment authorized by policy-engine (policy tier automatic_containment); action quarantine_vlan
2026-03-17 21:14:40.589	incident	sentinel	Incident INC-2026-0042 opened

2026-03-17 21:14:41.000	containment	sentinel	SW01 Gi1/0/23: switchport access vlan 20 → 50 (workflow WF-2026-0042-01)
2026-03-17 21:14:41.050	telemetry	telemetry	Telemetry from 10.20.0.47 no longer observed on monitored VLANs (23 further probe flows not seen)
2026-03-17 21:14:41.125	verification	sentinel	Verification succeeded: Access Mode VLAN: 50
2026-03-17 21:14:41.464	evidence	sentinel	Evidence bundle INC-2026-0042 finalized (895 related connections, 254 DNS records)
2026-03-17 21:49:31.284	resolution	tlee	Containment released by tlee: Endpoint re-imaged and cleared by IT; returning to production VLAN.

INC-2026-0041 · Rogue DHCP — VENDOR-LT-01 high resolved

Chain: `10.40.0.104 → 3C:D9:2B:71:31:04 → SW01 → Gi1/0/31 → VLAN 40 → shutdown` · opened 2026-03-17T11:23:47.191Z · closed 2026-03-17T11:47:20.288Z by tlee

TIME (UTC)	PHASE	ACTOR	EVENT
2026-03-17 11:23:44.106	telemetry	telemetry	DHCP snooping dropping DHCPOFFER from 3cd9.2b71.3104 on untrusted port Gi1/0/31 (client CONF-A-TV01 broadcast DHCPDISCOVER)
2026-03-17 11:23:44.106	switch	SW01	SW01 %DHCP_SNOOPING-5-DHCP_SNOOPING_UNTRUSTED_PORT: DHCP_SNOOPING drop message on untrusted port, message type: DHCPOFFER, MAC sa: 3cd9.2b71.3104
2026-03-17 11:23:44.131	alert	suricata	Suricata sid 1000040 (high): SENTINELSWITCH ROGUE DHCP server message from non-authorized host
2026-03-17 11:23:44.612	switch	SW01	SW01 %DHCP_SNOOPING-5-DHCP_SNOOPING_UNTRUSTED_PORT: DHCP_SNOOPING drop message on untrusted port, message type: DHCPNAK, MAC sa: 3cd9.2b71.3104
2026-03-17 11:23:45.130	switch	SW01	SW01 %DHCP_SNOOPING-5-DHCP_SNOOPING_UNTRUSTED_PORT: DHCP_SNOOPING drop message on untrusted port, message type: DHCPOFFER, MAC sa: 3cd9.2b71.3104
2026-03-17 11:23:45.655	switch	SW01	SW01 %DHCP_SNOOPING-5-DHCP_SNOOPING_UNTRUSTED_PORT: DHCP_SNOOPING drop message on untrusted port, message type: DHCPNAK, MAC sa: 3cd9.2b71.3104
2026-03-17 11:23:46.170	switch	SW01	SW01 %DHCP_SNOOPING-5-DHCP_SNOOPING_UNTRUSTED_PORT: DHCP_SNOOPING drop message on untrusted port, message type: DHCPOFFER, MAC sa: 3cd9.2b71.3104
2026-03-17 11:23:47.191	detection	sentinel	Rule rogue-dhcp matched: severity high, confidence 0.95 (base 0.80; repeated_drops+0.08, suricata_offer+0.07)
2026-03-17 11:23:47.260	correlation	sentinel	Resolving actor 3C:D9:2B:71:31:04 through DHCP snooping, MAC table and inventory
2026-03-17 11:23:47.399	correlation	sentinel	dhcp_binding [resolved] 3C:D9:2B:71:31:04 -> 10.40.0.104 (VLAN 40, Gi1/0/31)
2026-03-17 11:23:47.618	correlation	sentinel	mac_table [resolved] 3C:D9:2B:71:31:04 -> SW01 Gi1/0/31 (VLAN 40, DYNAMIC)
2026-03-17 11:23:47.618	correlation	sentinel	interface [resolved] Gi1/0/31 'VENDOR DROP (guest fallback)' mode=access access_vlan=20 oper_vlan=40 class=conference status=connected
2026-03-17 11:23:47.699	correlation	sentinel	identity [resolved] VENDOR-LT-01 (guest-laptop, HP, Windows 11 Home) criticality=low
2026-03-17 11:23:47.848	policy	sentinel	Disposition automatic_containment: Automatic containment approved (severity=high confidence=0.95); action=port_shutdown
2026-03-17 11:23:47.850	authorization	sentinel	Containment authorized by policy-engine (policy tier automatic_containment); action port_shutdown
2026-03-17 11:23:47.852	incident	sentinel	Incident INC-2026-0041 opened

2026-03-17 11:23:48.244	containment	sentinel	SW01 Gi1/0/31: administrative shutdown (workflow WF-2026-0041-04) — SentinelSwitch administrative shutdown
2026-03-17 11:23:48.366	verification	sentinel	Verification succeeded: admin down, no MAC addresses learned
2026-03-17 11:23:48.699	evidence	sentinel	Evidence bundle INC-2026-0041 finalized (8 related connections, 0 DNS records)
2026-03-17 11:47:00.000	resolution	analyst	Offending device physically removed after inspection
2026-03-17 11:47:20.288	resolution	tlee	Containment released by tlee: Vendor disabled connection sharing and left site. Port restored to service.

INC-2026-0040 · Unauthorized Endpoint — unknown device DC:A6:32:19:77:C1 high resolved

Chain: `DC:A6:32:19:77:C1 → SW01 → Gi1/0/17 → VLAN 20 → shutdown` · opened 2026-03-17T09:41:05.573Z · closed 2026-03-17T09:58:30.514Z by tlee

TIME (UTC)	PHASE	ACTOR	EVENT
2026-03-17 09:41:05.112	telemetry	telemetry	link up on Gi1/0/17: unregistered device dca6.3219.77c1 (Raspberry Pi Foundation OUI)
2026-03-17 09:41:05.418	aaa	telemetry	MAB Access-Reject for dca6.3219.77c1 on Gi1/0/17: MAC not in endpoint database
2026-03-17 09:41:05.418	aaa	freeradius	RADIUS Access-Reject (MAB) for DC:A6:32:19:77:C1 on Gi1/0/17: MAC not found in endpoint database
2026-03-17 09:41:05.444	switch	SW01	SW01 %MAB-5-FAIL: Authentication failed for client (dca6.3219.77c1) on Interface Gi1/0/17 AuditSessionID 0A0A0A02000000110019D48E
2026-03-17 09:41:05.573	detection	sentinel	Rule unauthorized-endpoint matched: severity high, confidence 0.96 (base 0.70; corporate_port+0.20, unknown_mac+0.06)
2026-03-17 09:41:05.644	correlation	sentinel	Resolving actor DC:A6:32:19:77:C1 through DHCP snooping, MAC table and inventory
2026-03-17 09:41:05.775	correlation	sentinel	dhcp_binding [missing] No binding for this MAC (expected when the port never authorised or the device is static)
2026-03-17 09:41:05.971	correlation	sentinel	mac_table [fallback] No MAC table entry for DC:A6:32:19:77:C1 (port not authorised); interface Gi1/0/17 taken from RADIUS NAS-Port-Id
2026-03-17 09:41:05.971	correlation	sentinel	interface [resolved] Gi1/0/17 'OPS SPARE DROP (corporate asset only)' mode=access access_vlan=20 oper_vlan=20 class=workstation status=connected
2026-03-17 09:41:06.048	correlation	sentinel	identity [missing] DC:A6:32:19:77:C1 is not in the asset inventory (OUI vendor: Raspberry Pi Foundation)
2026-03-17 09:41:06.200	policy	sentinel	Disposition automatic_containment: Automatic containment approved (severity=high confidence=0.96); action=port_shutdown
2026-03-17 09:41:06.202	authorization	sentinel	Containment authorized by policy-engine (policy tier automatic_containment); action port_shutdown
2026-03-17 09:41:06.204	incident	sentinel	Incident INC-2026-0040 opened
2026-03-17 09:41:06.601	containment	sentinel	SW01 Gi1/0/17: administrative shutdown (workflow WF-2026-0040-02) — SentinelSwitch administrative shutdown
2026-03-17 09:41:06.718	verification	sentinel	Verification succeeded: admin down, no MAC addresses learned
2026-03-17 09:41:07.039	evidence	sentinel	Evidence bundle INC-2026-0040 finalized (0 related connections, 0 DNS records)
2026-03-17 09:52:30.000	resolution	analyst	Offending device physically removed after inspection

2026-03-17 09:58:30.514	resolution	tlee	Containment released by tlee: Unregistered contractor single-board computer removed after physical inspection (ticket SVC-2026-0311). Port restored to service.

INC-2026-0039 · STP/BPDU Violation — CONF-A-TV01 high resolved

Chain: SW01 → Gi1/0/30 → VLAN 40 → shutdown · opened 2026-03-17T09:12:44.447Z · closed 2026-03-17T09:35:10.772Z by tlee

TIME (UTC)	PHASE	ACTOR	EVENT
2026-03-17 09:12:44.271	telemetry	telemetry	BPDU received on access port Gi1/0/30 (unmanaged switch inserted in front of CONF-A-TV01); BPDU Guard err-disabled the port
2026-03-17 09:12:44.271	switch	SW01	SW01 %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi1/0/30 with BPDU Guard enabled. Disabling port.
2026-03-17 09:12:44.276	switch	SW01	SW01 %PM-4-ERR_DISABLE: bpduguard error detected on Gi1/0/30, putting Gi1/0/30 in err-disable state
2026-03-17 09:12:44.447	detection	sentinel	Rule bpdu-violation matched: severity high, confidence 0.96 (base 0.90; errdisable_confirmed+0.06)
2026-03-17 09:12:44.513	correlation	sentinel	Resolving actor Gi1/0/30 through DHCP snooping, MAC table and inventory
2026-03-17 09:12:44.650	correlation	sentinel	dhcp_binding [skipped] No IP or MAC to look up
2026-03-17 09:12:44.858	correlation	sentinel	mac_table [fallback] No MAC table entry for unknown MAC (port not authorised); interface Gi1/0/30 taken from syslog interface field
2026-03-17 09:12:44.858	correlation	sentinel	interface [resolved] Gi1/0/30 'CONF-A DISPLAY (guest fallback)' mode=access access_vlan=20 oper_vlan=40 class=conference status=err-disabled
2026-03-17 09:12:44.937	correlation	sentinel	identity [fallback] No MAC in the event; CONF-A-TV01 is the endpoint assigned to Gi1/0/30 (display, criticality=low)
2026-03-17 09:12:45.081	policy	sentinel	Disposition automatic_containment: Automatic containment approved (severity=high confidence=0.96); action=port_shutdown
2026-03-17 09:12:45.083	authorization	sentinel	Containment authorized by policy-engine (policy tier automatic_containment); action port_shutdown
2026-03-17 09:12:45.085	incident	sentinel	Incident INC-2026-0039 opened
2026-03-17 09:12:45.469	containment	sentinel	SW01 Gi1/0/30: administrative shutdown (workflow WF-2026-0039-01) — BPDU Guard err-disable (switch-native) + SentinelSwitch administrative hold
2026-03-17 09:12:45.588	verification	sentinel	Verification succeeded: admin down, no MAC addresses learned
2026-03-17 09:12:45.915	evidence	sentinel	Evidence bundle INC-2026-0039 finalized (0 related connections, 0 DNS records)
2026-03-17 09:34:40.000	resolution	analyst	Offending device physically removed after inspection
2026-03-17 09:35:10.772	resolution	tlee	Containment released by tlee: Unmanaged switch removed from conference room; display reconnected directly. Port recovered with shutdown / no shutdown.

7. Containment authorization and verification

WORKFLOW	INCIDENT	ACTION	AUTHORIZATION	AUTHORIZED AT	APPLIED	VERIFIED	READ-BACK	STATUS
WF-2026-0042-01	INC-2026-0042	quarantine_vlan Gi1/0/23	automatic · policy-engine policy tier automatic_containment	2026-03-17 21:14:40.585	2026-03-17 21:14:41.000	2026-03-17 21:14:41.125	PASS	restored

WF-2026-0041-04	INC-2026-0041	port_shutdown Gi1/0/31	automatic · policy-engine policy tier automatic_containment	2026-03-17 11:23:47.848	2026-03-17 11:23:48.244	2026-03-17 11:23:48.366	PASS	restored
WF-2026-0040-02	INC-2026-0040	port_shutdown Gi1/0/17	automatic · policy-engine policy tier automatic_containment	2026-03-17 09:41:06.200	2026-03-17 09:41:06.601	2026-03-17 09:41:06.718	PASS	restored
WF-2026-0039-01	INC-2026-0039	port_shutdown Gi1/0/30	automatic · policy-engine policy tier automatic_containment	2026-03-17 09:12:45.081	2026-03-17 09:12:45.469	2026-03-17 09:12:45.588	PASS	restored

Detections routed to manual review

DETECTION	TITLE	CONFIDENCE	POLICY DISPOSITION	GUARDS	AUTHORIZATION	OUTCOME
DET-20260317T20371618-PORT	Critical Asset Scan — SRV-GIT01	0.96	manual_review	critical_assets	rejected · tlee	closed: Containment declined: critical_assets; investigated and closed as benign after review.
DET-20260317T14192804-DNS-	DNS Anomaly — FIN-WS-014	0.8	manual_review	–	rejected · tlee	closed: Endpoint-management agent on FIN-WS-014 was querying decommissioned telemetry hostnames after a failed update. Agent updated; queries stopped. Closed as benign.
DET-20260317T13051421-PORT	Port Scan — IT-LT-037	0.61	monitor	–	rejected · tlee	closed: Authorized vulnerability scan by IT-LT-037 under change CHG-2026-118. Logged only.
DET-20260317T12365740-MAC-	MAC Address Change — HR-WS-016	0.87	manual_review	–	rejected · tlee	closed: HR user connected a personal laptop through the docking station on HR-WS-016's drop. Frames were restricted by port security; user counseled, personal device removed. Closed.
DET-20260317T10272329-ARP-	ARP Spoofing Attempt — GUEST-VISITOR-01	0.78	manual_review	–	rejected · tlee	closed: Guest laptop VPN client emitted stale gratuitous ARP for its previous network. DAI dropped all three frames; no further activity. Closed as benign.

8. Outstanding remediation

No outstanding remediation.

Remediation history

FINDING	TIME	EVENT	ACTOR	DETAIL
FND-CONFIG-BASELINE-Gi1-0-22	2026-03-17T21:14:00.000Z	opened	sentinel-assurance	FAIL in run ASR-20260317T211400-baseline-seed (baseline-seed)
FND-CONFIG-BASELINE-Gi1-0-22	2026-03-17T21:15:31.284Z	remediation_requested	tlee	SentinelSwitch 2.0 demo remediation
FND-CONFIG-BASELINE-Gi1-0-22	2026-03-17T21:15:35.484Z	remediation_executed	sentinel	configure terminal / interface GigabitEthernet1/0/22 / description MKT-WS-022 / end / write memory
FND-CONFIG-BASELINE-Gi1-0-22	2026-03-17T21:15:47.484Z	verified	sentinel-assurance	remediation verified by run ASR-20260317T211547-remediation-FND-CONFIG-BASELINE-Gi1-0-22
FND-CONFIG-BASELINE-Gi1-0-35	2026-03-17T21:14:00.000Z	opened	sentinel-assurance	FAIL in run ASR-20260317T211400-baseline-seed (baseline-seed)
FND-CONFIG-BASELINE-Gi1-0-35	2026-03-17T21:16:31.284Z	remediation_requested	tlee	SentinelSwitch 2.0 demo remediation
FND-CONFIG-BASELINE-Gi1-0-35	2026-03-17T21:16:35.484Z	remediation_executed	sentinel	configure terminal / interface GigabitEthernet1/0/35 / storm-control broadcast level 1.00 / storm-control multicast level 1.00 / storm-control action trap / end / write memory
FND-CONFIG-BASELINE-Gi1-0-35	2026-03-17T21:16:47.484Z	verified	sentinel-assurance	remediation verified by run ASR-20260317T211647-remediation-FND-CONFIG-BASELINE-Gi1-0-35
FND-ENDPOINT-IDENTITY-Gi1-0-30	2026-03-17T21:14:00.000Z	opened	sentinel-assurance	STALE in run ASR-20260317T211400-baseline-seed (baseline-seed)
FND-ENDPOINT-IDENTITY-Gi1-0-30	2026-03-17T21:17:31.284Z	remediation_requested	tlee	SentinelSwitch 2.0 demo remediation
FND-ENDPOINT-IDENTITY-Gi1-0-30	2026-03-17T21:17:35.484Z	remediation_executed	sentinel	clear authentication sessions interface GigabitEthernet1/0/30
FND-ENDPOINT-IDENTITY-Gi1-0-30	2026-03-17T21:17:47.484Z	verified	sentinel-assurance	remediation verified by run ASR-20260317T211747-remediation-FND-ENDPOINT-IDENTITY-Gi1-0-30
FND-SENSOR-COVERAGE-vlan-50	2026-03-17T21:14:00.000Z	opened	sentinel-assurance	FAIL in run ASR-20260317T211400-baseline-seed (baseline-seed)
FND-SENSOR-COVERAGE-vlan-50	2026-03-17T21:18:31.284Z	remediation_requested	tlee	SentinelSwitch 2.0 demo remediation
FND-SENSOR-COVERAGE-vlan-50	2026-03-17T21:18:35.484Z	remediation_executed	sentinel	configure terminal / monitor session 1 source vlan 20 , 30 , 40 , 50 both / end / write memory
FND-SENSOR-COVERAGE-vlan-50	2026-03-17T21:18:47.484Z	verified	sentinel-assurance	remediation verified by run ASR-20260317T211847-remediation-FND-SENSOR-COVERAGE-vlan-50
FND-UNUSED-PORTS-Gi1-0-41	2026-03-17T21:14:00.000Z	opened	sentinel-assurance	FAIL in run ASR-20260317T211400-baseline-seed (baseline-seed)

FND-UNUSED-PORTS-Gi1-0-41	2026-03-17T21:19:31.284Z	remediation_requested	tlee	SentinelSwitch 2.0 demo remediation
FND-UNUSED-PORTS-Gi1-0-41	2026-03-17T21:19:35.484Z	remediation_executed	sentinel	configure terminal / interface GigabitEthernet1/0/41 / shutdown / switchport access vlan 999 / end / write memory
FND-UNUSED-PORTS-Gi1-0-41	2026-03-17T21:19:47.484Z	verified	sentinel-assurance	remediation verified by run ASR-20260317T211947-remediation-FND-UNUSED-PORTS-Gi1-0-41

9. Framework mapping (NIST CSF 2.0)

Limited mapping of simulated checks to CSF 2.0 subcategories for orientation only; not a framework assessment or certification.

SUBCATEGORY	TEXT	CONTROLS	PASS	FAIL	UNKNOWN	STALE
DE.AE-03	Information is correlated from multiple sources	CTL-EVIDENCE-COMPLETE	4	0	0	0
DE.CM-01	Networks and network services are monitored to find potentially adverse events	CTL-SENSOR-COVERAGE, CTL-TELEMETRY-FRESH	7	0	1	0
ID.AM-01	Inventories of hardware managed by the organization are maintained	CTL-ENDPOINT-IDENTITY	28	0	0	0
ID.AM-03	Representations of the organization's authorized network communication and internal and external network data flows are maintained	CTL-SENSOR-COVERAGE, CTL-VLAN-INTENT	40	0	0	0
PR.AA-03	Users, services, and hardware are authenticated	CTL-DOT1X-ENFORCED, CTL-ENDPOINT-IDENTITY, CTL-PORT-SECURITY	90	0	0	0
PR.IR-01	Networks and environments are protected from unauthorized logical access and usage	CTL-DOT1X-ENFORCED, CTL-PORT-SECURITY, CTL-TRUST-BOUNDARY, CTL-UNUSED-PORTS, CTL-VLAN-INTENT	111	0	0	0
PR.PS-01	Configuration management practices are established and applied	CTL-CONFIG-BASELINE, CTL-UNUSED-PORTS	62	0	0	0
RC.RP-01	The recovery portion of the incident response plan is executed once initiated from the incident response process	CTL-RECOVERY-VERIFIED	4	0	0	0
RC.RP-04	Critical mission functions and cybersecurity risk management are considered to establish post-incident operational norms	CTL-RECOVERY-VERIFIED	4	0	0	0
RS.AN-06	Actions performed during an investigation are recorded, and the records' integrity and provenance are preserved	CTL-EVIDENCE-COMPLETE, CTL-RESPONSE-AUTHORIZED	8	0	0	0
RS.MA-01	The incident response plan is executed in coordination with relevant third parties once an incident is declared	CTL-RESPONSE-AUTHORIZED	4	0	0	0

10. Limitations and reproducibility

- The switch is a digital twin; no commands were sent to physical hardware.
- Telemetry is synthesized from a deterministic traffic model; Zeek and Suricata were not executed.
- Suricata liveness cannot be assessed because the dataset has no EVE stats events.
- Timestamps are simulated-clock values, not wall-clock execution times.

Reproduce: `make seed ; make demo-2 ; make report`. Dataset manifest: `data/generated/manifest.json`. Validation harness: 9/9 scenarios, MTTD 10.446 s, MTTC 1.214 s (simulated clock).

Report rendered from persisted records at simulated clock 2026-03-17T21:49:31.284Z; no wall-clock timestamps are embedded so repeated exports of the same state are identical.